

Phase-by-Phase FNO and PINO Training for WaveQLab2D

WaveQLab2D Simulation Pipeline

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1 Purpose and experimental contract

The objective is to learn the map from a Gaussian source position and query time to the full particle-velocity field,

$$\mathcal{G} : (x_0, y_0, t) \mapsto \mathbf{v}(x, y, t) = (v_x(x, y, t), v_y(x, y, t)),$$

on the fixed 10×10 km objective window. The supervised FNO and PINO use the same raw simulations, prepared tensors, normalization, architecture, optimizer family, validation cases, and test cases. PINO is warm-started from FNO and adds a wave-equation residual. This isolates the effect of physics regularization.

The pilot contains 500 simulations: 350 training, 50 validation, 50 ID test, and 50 far-OOD test cases. Training source offsets are in $[-3, 3]$ km; far-OOD offsets are in $[-5, -4) \cup (4, 5]$ km. Designs at 500, 1000, 1500, and 2000 samples are nested, so learning curves do not change earlier examples.

2 Governing equations

For homogeneous isotropic linear elasticity,

$$\rho \partial_t \mathbf{v} = \nabla \cdot \boldsymbol{\sigma} + \mathbf{f}, \quad (1)$$

$$\partial_t \boldsymbol{\sigma} = \lambda(\nabla \cdot \mathbf{v}) \mathbf{I} + \mu(\nabla \mathbf{v} + \nabla \mathbf{v}^T), \quad (2)$$

where ρ is density and λ, μ are Lamé parameters. Eliminating stress away from the source gives the velocity Navier equation

$$\partial_{tt} \mathbf{v} = c_s^2 \nabla^2 \mathbf{v} + (c_p^2 - c_s^2) \nabla(\nabla \cdot \mathbf{v}), \quad (3)$$

with $c_p^2 = (\lambda + 2\mu)/\rho$ and $c_s^2 = \mu/\rho$.

In components,

$$R_x = v_{x,tt} - c_p^2 v_{x,xx} - c_s^2 v_{x,yy} - (c_p^2 - c_s^2) v_{y,xy}, \quad (4)$$

$$R_y = v_{y,tt} - c_s^2 v_{y,xx} - c_p^2 v_{y,yy} - (c_p^2 - c_s^2) v_{x,xy}. \quad (5)$$

The PINO physics loss is the mean squared nondimensional residual outside a source mask. Centered second differences use the saved spacing $\Delta t_s = 0.02$ s and $\Delta x = \Delta y = 0.1$ km. The source mask radius is 0.5 km, exceeding twice the solver Gaussian standard deviation $\sigma = 2\Delta x = 0.2$ km.

3 Fourier Neural Operator

Let $a(x, y)$ denote the input-channel field. A spectral layer applies

$$z_{\ell+1} = \phi(W_\ell z_\ell + \mathcal{F}^{-1}(R_\ell \mathcal{F} z_\ell)), \quad (6)$$

where W_ℓ is a learned pointwise map and R_ℓ contains learned complex weights for a truncated set of Fourier modes. The retained low modes efficiently represent propagating wave structure while the local 1×1 convolution restores local coupling.

The model inputs are: the spatial Gaussian source, source multiplied by its temporal envelope, normalized x and y , distance to the objective boundary, normalized query time, normalized y_0 , and three sine/cosine Fourier pairs in time. There are 13 channels and two output channels.

4 Losses

The supervised loss combines pointwise, gradient, and spectral agreement:

$$\mathcal{L}_{\text{data}} = \text{MSE}(\hat{v}, v) + 0.05 \text{MAE}(\hat{v}, v) + 0.05(\text{MAE}(D_x \hat{v}, D_x v) + \text{MAE}(D_y \hat{v}, D_y v)) + 0.01 \text{MAE}(|\mathcal{F} \hat{v}|, |\mathcal{F} v|). \quad (7)$$

PINO minimizes

$$\mathcal{L}_{\text{PINO}} = \mathcal{L}_{\text{data}} + \gamma \mathcal{L}_{\text{physics}}, \quad \mathcal{L}_{\text{physics}} = \frac{1}{|\Omega_m|} \sum_{\Omega_m} \left\| \frac{\mathbf{R}}{v_{\text{scale}}/\Delta t_s^2} \right\|_2^2. \quad (8)$$

The default $\gamma = 0.01$ follows the earlier pilot. It must be tuned only on the validation split; recommended values are $10^{-3}, 3 \times 10^{-3}, 10^{-2}, 3 \times 10^{-2}$.

5 Phase algorithms

5.1 Phase 01: integrity and preparation

`phase_01.py` finds exactly one finalized raw HDF5 archive per manifest row, merges the left/right blocks while removing the duplicate interface point, selects only v_x, v_y , transposes to $[t, x, y, 2]$, checks shape $[250, 101, 101, 2]$ and finite values, then writes a chunked compressed prepared HDF5 file.

```
left = source["DomainOutput_l"][:-1, :, :, :2]
right = source["DomainOutput_r"][:, :, :, :2]
velocity = np.concatenate((left, right), axis=0).transpose(2,0,1,3)
```

5.2 Phase 02: leakage-free normalization

`phase_02.py` reads training cases only. Per-component maximum magnitude and RMS are accumulated in time chunks. The scale is

$$s_k = \max(\max |v_k|, 5 \text{ RMS}(v_k)).$$

Coordinates, saved times, spacings, scales, and field order are stored in `normalization.json`. Validation and test data never influence this file.

5.3 Phase 03: baseline FNO

Each optimization step samples cases and query frames, constructs the conditioning channels, scales the target velocities, evaluates the FNO, and applies the composite supervised loss. AdamW, cosine learning-rate decay, unit gradient clipping, and validation relative L_2 checkpoint selection are used.

5.4 Phase 04: bounded hyperparameter tuning

`phase_04.py` searches width $\{24, 32, 48\}$, modes $\{12, 16, 20\}$, and learning rate $\{5 \times 10^{-4}, 10^{-3}\}$. Short trials rank configurations by validation relative L_2 . The winning configuration should then be retrained with the full Phase 03 budget and at least three seeds. The test set is not consulted.

5.5 Phase 05: PINO fine-tuning

Three consecutive query times are passed through the same FNO. Their predicted fields approximate v_{tt} , while spatial differences approximate the derivatives in Eq. (3). Fine-tuning starts from the best FNO checkpoint with a smaller learning rate 10^{-4} to preserve the learned data manifold.

```
predicted = model(triplet_inputs)
data = supervised_loss(predicted_current, truth_current)
physics = navier(predicted, cases, metadata, scale)
loss = data + physics_weight * physics
```

5.6 Phase 06: matched evaluation

Both checkpoints predict every saved test frame. The report separates ID and far-ODD cases and records global relative L_2 , per-case relative L_2 , maximum absolute error, median/p90 error, total inference time, and milliseconds per frame. Evaluation uses identical batches and normalization for FNO and PINO.

6 Parameter and hyperparameter rationale

Parameter	Default	Rationale
Spatial modes	16	Resolves wavelengths represented on a 101^2 grid without the cost of retaining all modes.
Width	32	Moderate H100 memory use; width search tests under/over-capacity.
Layers	4	Repeated global/local mixing with stable optimization.
Batch size	4 FNO, 2 PINO	PINO evaluates three times per sampled case.
FNO learning rate	10^{-3}	AdamW baseline; searched against 5×10^{-4} .
PINO learning rate	10^{-4}	Conservative warm-start fine-tuning.
Weight decay	10^{-4}	Limits spectral-weight overfit.
Physics weight	10^{-2}	Previous pilot baseline; tune logarithmically.
Source mask	0.5 km	Excludes the forced region where homogeneous Eq. (3) is invalid.
Checkpoint metric	validation relative L_2	Directly targets physical field accuracy.

7 Accuracy-first tuning protocol

First verify overfitting on a tiny subset; failure indicates a code or conditioning problem. Next tune FNO capacity and optimizer settings using all validation cases. Retrain the winner with three seeds. Tune PINO physics weight from weak to strong, rejecting settings that reduce residual but degrade validation field error. Finally evaluate once on ID and far-OOD test groups. For the nested 500/1000/1500/2000 study, preserve hyperparameter-search rules and report both single-seed and ensemble learning curves.

8 Reproducible execution

```
python phase_01.py DATASET
python phase_02.py DATASET
python phase_03.py DATASET --output-dir DATASET/models/fno
python phase_04.py DATASET --output-dir DATASET/models/search
python phase_05.py DATASET DATASET/models/fno/best.pt \
  --output-dir DATASET/models/pino
python phase_06.py DATASET --fno DATASET/models/fno/best.pt \
  --pino DATASET/models/pino/best.pt --output-dir DATASET/evaluation
```